Level 2

Hackathon – Fundamentals of Web Development

Student Details

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1. Problem Statement

To develop a user-friendly web application using HTML, CSS, and JavaScript to display and search college clubs, register students with validation, manage club memberships, display members, and publish upcoming events. Local Storage is used to save membership and event details.

2. Objectives

To display college clubs dynamically.

To search and filter clubs easily.

To display club information and members.

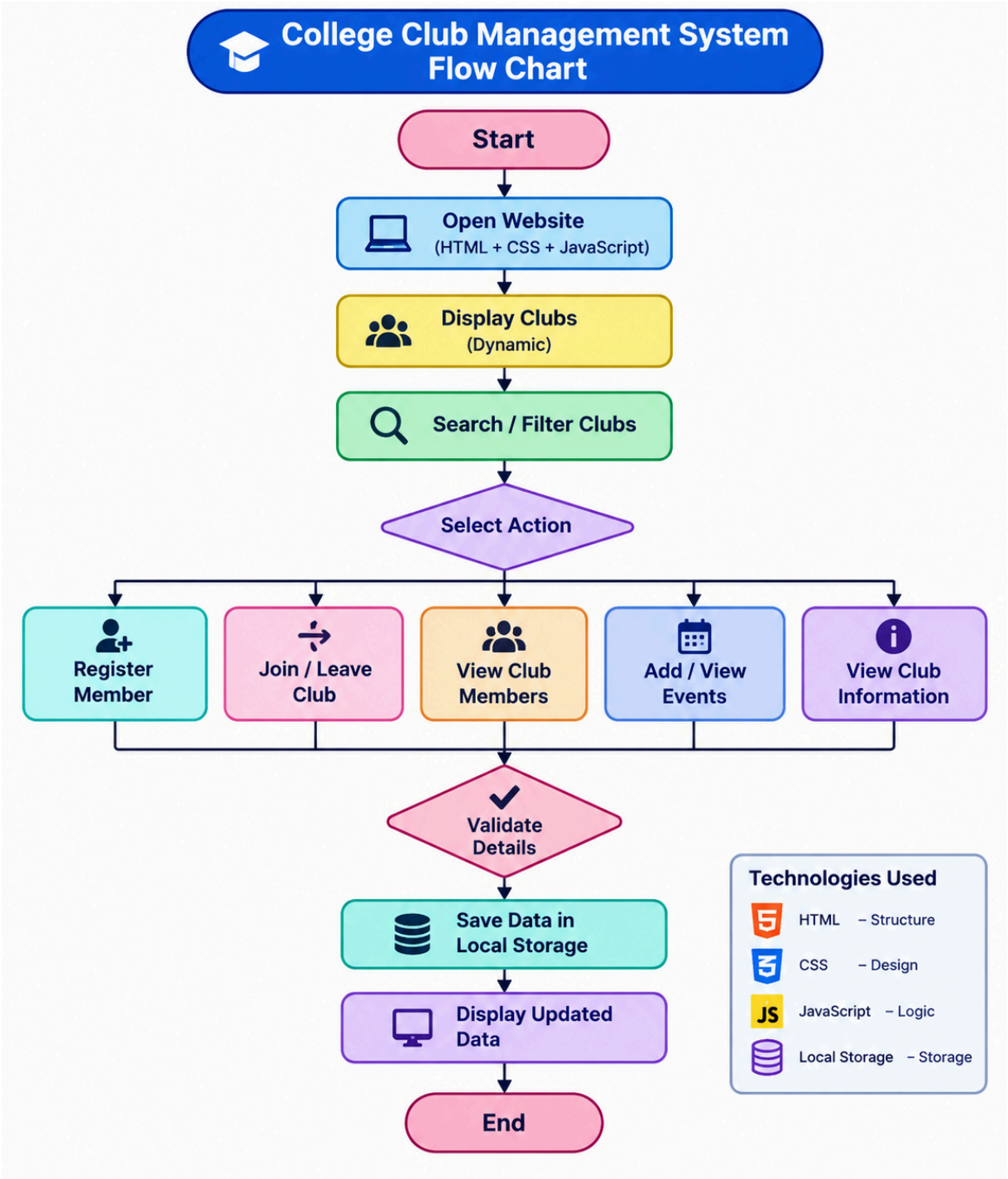
To register students with validation.

To manage joining and leaving clubs.

To add and display event announcements.
To show upcoming club events.

To store membership details using Local Storage.

3. Architecture



4. Practical Implementation

Display college clubs and their details dynamically.

Allow students to register, join clubs, and view events.

Store membership and event details using Local Storage.

4.1 Features Implemented

Dynamic display, search, and filter of college clubs.

Student registration, join/leave clubs, and member details.

Event announcements, upcoming events, and Local Storage.

4.2 Source Code

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>Campus Club Hub</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<header>
<h1>🎓 Campus Club Hub</h1>
<p>Discover Clubs | Join Communities | Explore Events</p>
</header>

<main class="container">

<div class="stats">
<div class="stat"><h2 id="clubCount">0</h2>Available Clubs</div>
<div class="stat"><h2 id="memberCount">0</h2>Memberships</div>
<div class="stat"><h2 id="eventCount">0</h2>Upcoming Events</div>
</div>

<section class="panel">
<h2>🔎 Explore Clubs</h2>
<input id="search" placeholder="Search clubs...">
<select id="categoryFilter">
<option value="All">All Categories</option>
<option>Technology</option>
<option>Arts</option>
<option>Sports</option>
```

<option>Social Service</option>
<option>Literature</option>
</select>
</section>

<section class="panel" id="registration">
<h2>📝 Student Registration</h2>
<form id="registrationForm">
<input id="studentName" placeholder="Full Name" minlength="3" required>
<input id="studentId" placeholder="Student ID" required>
<input id="studentEmail" type="email" placeholder="Email" required>
<input id="studentPhone" type="tel" pattern="[0-9]{10}" placeholder="10-digit Phone" required>
<select id="joinClub" required></select>
<button type="submit">Register & Join Club</button>
</form>
<p id="message" class="notice" hidden></p>
</section>

<section>
<h2>🌟 Available Clubs</h2>

</section>

<section class="panel">
<h2>👥 Club Memberships</h2>

</section>

<section class="panel">
<h2>📢 Publish Event</h2>
<form id="eventForm">
<input id="eventTitle" placeholder="Event Title" required>
<select id="eventClub" required></select>

<input id="eventDate" type="date" required>
<input id="eventVenue" placeholder="Event Venue" required>
<textarea id="eventDescription" placeholder="Event Description" required></textarea>
<button type="submit">Publish Event</button>
</form>
</section>

<section class="panel">
<h2>📅 Upcoming Events</h2>

</section>

</main>

<script src="script.js"></script>
</body>
</html>

CSS

```
* { box-sizing: border-box; }

body {
  margin: 0;
  font-family: Arial;
  background: #f1f5f9;
  color: #1e293b;
}

header {
  background: linear-gradient(120deg,#312e81,#7c3aed);
  color: white;
  text-align: center;
  padding: 25px 12px;
}

.container {
```

```
  max-width: 1100px;
  margin: auto;
  padding: 16px;
}

.stats, .clubs, .events {
  display: grid;
  grid-template-columns: repeat(auto-fit,minmax(220px,1fr));
  gap: 14px;
}

.panel, .card, .stat {
  background: white;
  padding: 16px;
  border-radius: 14px;
  box-shadow: 0 3px 12px #0001;
  margin-bottom: 16px;
}

.stat { text-align: center; }
.stat h2, .card h3 { color: #7c3aed; }

input, select, textarea, button {
  width: 100%;
  padding: 11px;
  margin: 6px 0;
  border: 1px solid #cbd5e1;
  border-radius: 8px;
  font-size: 15px;
}

button {
  background: #6d28d9;
  color: white;
  border: 0;
  font-weight: bold;
  cursor: pointer;
}

button:hover { opacity: .85; }
.secondary { background: #475569; }
.danger { background: #dc2626; }
```

```
.tag {
  background: #ede9fe;
  color: #5b21b6;
  padding: 5px 9px;
  border-radius: 15px;
  font-size: 12px;
}

.member-list, .notice {
  background: #f8fafc;
  padding: 10px;
  border-radius: 8px;
}
```

JavaScript

```
const defaultClubs = [
{id:1,name:"Coding Club",category:"Technology",icon:"💻",description:"Learn web development."},
{id:2,name:"Arts Club",category:"Arts",icon:"🎨",description:"Explore art and creativity."},
{id:3,name:"Sports Club",category:"Sports",icon:"🏆",description:"Join games and tournaments."},
{id:4,name:"Social Service Club",category:"Social Service",icon:"🤝",description:"Help people through volunteering."},
{id:5,name:"Literature Club",category:"Literature",icon:"📚",description:"Enjoy reading and writing."},
{id:6,name:"Robotics Club",category:"Technology",icon:"🤖",description:"Explore robotics and innovation."}
];

function loadData(key, fallback) {
  try {
    const data = JSON.parse(localStorage.getItem(key));
    return data === null ? fallback : data;
  } catch {
    return fallback;
  }
}

let clubs = loadData("campusClubs", defaultClubs);
let members = loadData("campusMembers", []);
let events = loadData("campusEvents", []);
```

```
function saveData() {
  localStorage.setItem("campusClubs", JSON.stringify(clubs));
  localStorage.setItem("campusMembers", JSON.stringify(members));
  localStorage.setItem("campusEvents", JSON.stringify(events));
}

function safe(value) {
  return String(value).replace(/[&<>"']/g, char => ({
    "&":"&amp;", "<":"&lt;", ">":"&gt;",
    '"':"&quot;", "'":"&#39;"
  }[char]));
}

function updateClubOptions() {
  const options = clubs.map(c =>
    `<option value="${c.id}">${safe(c.name)}</option>`
  ).join("");

  document.getElementById("joinClub").innerHTML =
    '<option value="">Select a Club</option>' + options;

  document.getElementById("eventClub").innerHTML =
    '<option value="">Select a Club</option>' + options;
}

function renderClubs() {
  const query = document.getElementById("search").value.toLowerCase();
  const category = document.getElementById("categoryFilter").value;

  const filtered = clubs.filter(c =>
    (c.name + " " + c.description + " " + c.category)
    .toLowerCase().includes(query) &&
    (category === "All" || c.category === category)
  );

  document.getElementById("clubList").innerHTML =
    filtered.map(c => {
      const count = members.filter(m => m.clubId === c.id).length;
      const joined = members.some(m => m.clubId === c.id);

      return `
      <article class="card">
```

```
    <h3>${c.icon} ${safe(c.name)}</h3>
    <span class="tag">${safe(c.category)}</span>
    <p>${safe(c.description)}</p>
    <p><b>Members:</b> ${count}</p>
    <button class="secondary" onclick="viewMembers(${c.id})">View
Members</button>
    ${joined
      ? `<button class="danger" onclick="leaveClub(${c.id})">Leave Club</button>`
      : `<button onclick="chooseClub(${c.id})">Join Club</button>`}
    <div id="memberList${c.id}" class="member-list" hidden></div>
    </article>`;
  }).join("") || "<p>No clubs found.</p>";

 document.getElementById("clubCount").textContent = clubs.length;
 document.getElementById("memberCount").textContent = members.length;
}

document.getElementById("search").addEventListener("input", renderClubs);
document.getElementById("categoryFilter").addEventListener("change",
renderClubs);

function chooseClub(id) {
 document.getElementById("joinClub").value = id;
 document.getElementById("registration").scrollIntoView({behavior:"smooth"});
}

document.getElementById("registrationForm").addEventListener("submit",
function(e) {
 e.preventDefault();

 const name = document.getElementById("studentName").value.trim();
 const studentId = document.getElementById("studentId").value.trim();
 const email = document.getElementById("studentEmail").value.trim();
 const phone = document.getElementById("studentPhone").value.trim();
 const clubId = Number(document.getElementById("joinClub").value);

 if (name.length < 3 || !/^[0-9]{10}$/.test(phone)) {
  alert("Enter a valid name and 10-digit phone number.");
  return;
 }

 if (members.some(m =>
  m.studentId.toLowerCase() === studentId.toLowerCase() &&
```

```
    m.clubId === clubId
  )) {
    alert("Already joined this club.");
    return;
  }

  members.push({
    id: Date.now(), name, studentId, email, phone, clubId,
    joinedOn: new Date().toISOString().slice(0,10)
  });

  saveData();
  this.reset();
  renderAll();
  showMessage("Registration successful!");
});

function leaveClub(clubId) {
  if (!confirm("Are you sure you want to leave?")) return;

  const index = members.findIndex(m => m.clubId === clubId);
  if (index === -1) return;

  members.splice(index, 1);
  saveData();
  renderAll();
  showMessage("You have left the club.");
}

function viewMembers(clubId) {
  const box = document.getElementById("memberList" + clubId);
  box.hidden = !box.hidden;
  if (box.hidden) return;

  const list = members.filter(m => m.clubId === clubId);

  box.innerHTML = "<b>Club Members</b>" +
    (list.map(m => `<p>👤 ${safe(m.name)} — ${safe(m.studentId)}</p>`).join("")
    || "<p>No members yet.</p>");
}

function renderMemberships() {
  document.getElementById("myMemberships").innerHTML =
```

```
  members.map(m => {
    const club = clubs.find(c => c.id === m.clubId);
    return `<div class="member-list">
    <b>${safe(m.name)}</b> — ${safe(club ? club.name : "Club")}
    <p>Student ID: ${safe(m.studentId)}</p>
    <small>Joined: ${safe(m.joinedOn)}</small>
    </div><br>`;
  }).join("") || "<p>No members registered yet.</p>";
}

document.getElementById("eventForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const date = document.getElementById("eventDate").value;
  const today = new Date().toISOString().slice(0,10);

  if (date < today) {
    alert("Choose today or a future date.");
    return;
  }

  events.push({
    id: Date.now(),
    title: document.getElementById("eventTitle").value.trim(),
    clubId: Number(document.getElementById("eventClub").value),
    date,
    venue: document.getElementById("eventVenue").value.trim(),
    description: document.getElementById("eventDescription").value.trim()
  });

  saveData();
  this.reset();
  renderAll();
  showMessage("Event published successfully!");
});

function renderEvents() {
  const today = new Date().toISOString().slice(0,10);
  const upcoming = events.filter(e => e.date >= today)
    .sort((a,b) => a.date.localeCompare(b.date));

  document.getElementById("eventCount").textContent = upcoming.length;
```

```
 document.getElementById("eventList").innerHTML =
  upcoming.map(e => {
    const club = clubs.find(c => c.id === e.clubId);
    return `<article class="card">
    <h3>📢 ${safe(e.title)}</h3>
    <span class="tag">${safe(club ? club.name : "Club")}</span>
    <p>📅 ${safe(e.date)}</p>
    <p>📍 ${safe(e.venue)}</p>
    <p>${safe(e.description)}</p>
    </article>`;
  }).join("") || "<p>No upcoming events available.</p>";
}

function showMessage(text) {
 const box = document.getElementById("message");
 box.textContent = text;
 box.hidden = false;
}

function renderAll() {
 updateClubOptions();
 renderClubs();
 renderMemberships();
 renderEvents();
}

renderAll();
```

4.3 Project Execution

Run the website using a web browser.

Search clubs, register students, and manage memberships.

Add events and save details using Local Storage.

5. Output Screenshots

Step 1: Display Clubs Dynamically

🌟 Available Clubs

💻 Coding Club

Technology

Learn HTML, CSS, JavaScript and build web projects.

Members: 1

View Members

Leave Club

🎨 Arts Club

Arts

Explore drawing, painting, music and creativity.

Members: 0

View Members

Step 2: Search and Filter Clubs

🔍 Explore Clubs

Search clubs...

Arts

📝 Student Registration

Full Name

Student ID / Roll Number

Email Address

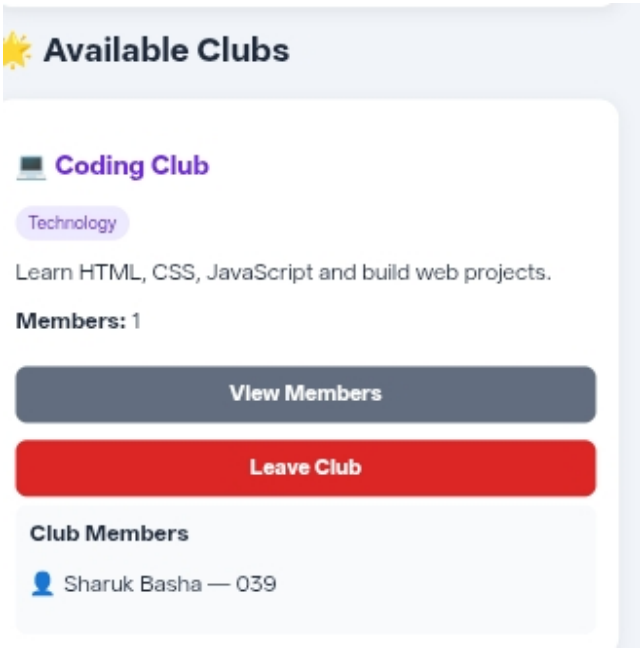
10-digit Phone Number

Sports Club

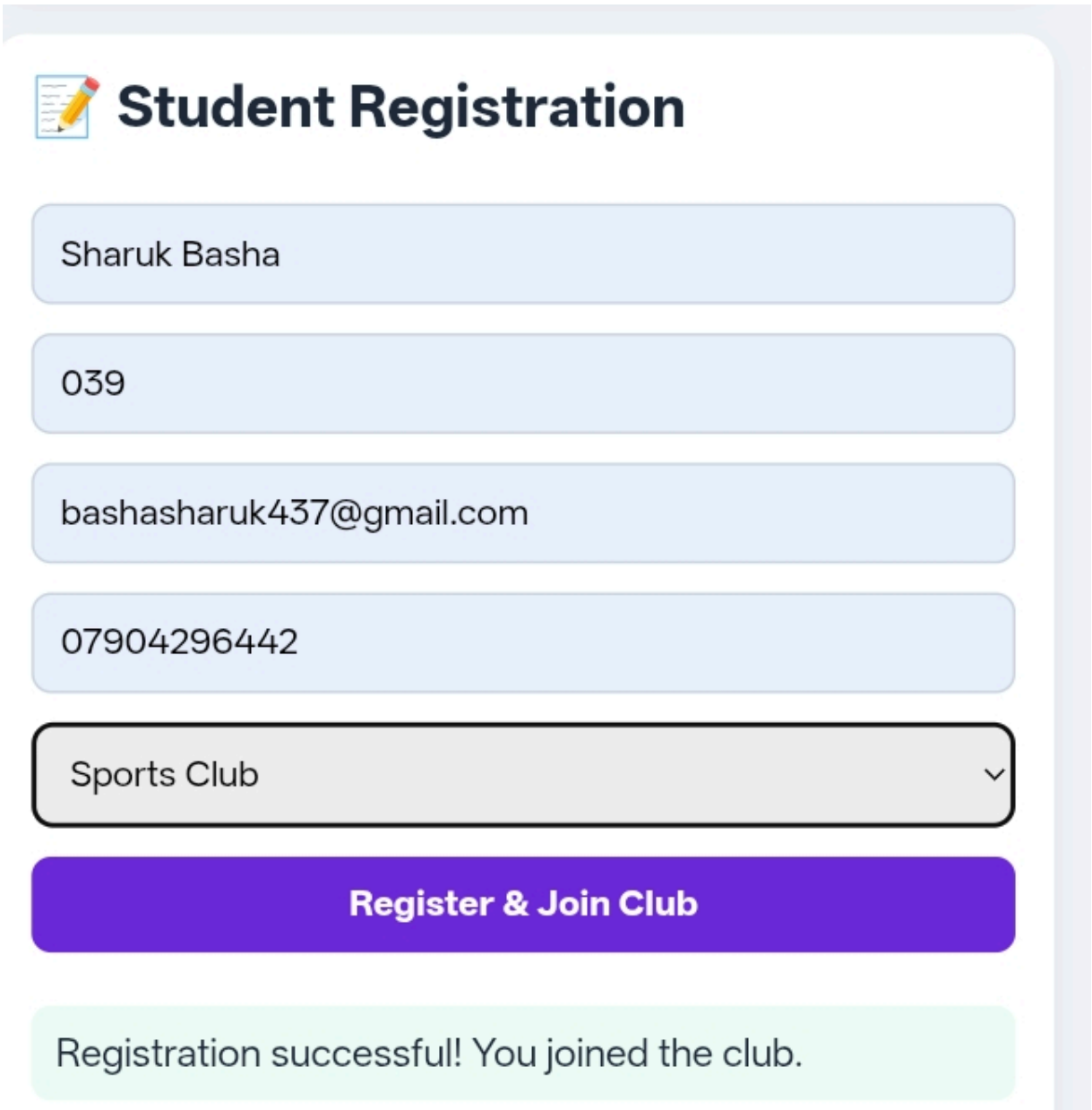
Register & Join Club

🌟 Available Clubs

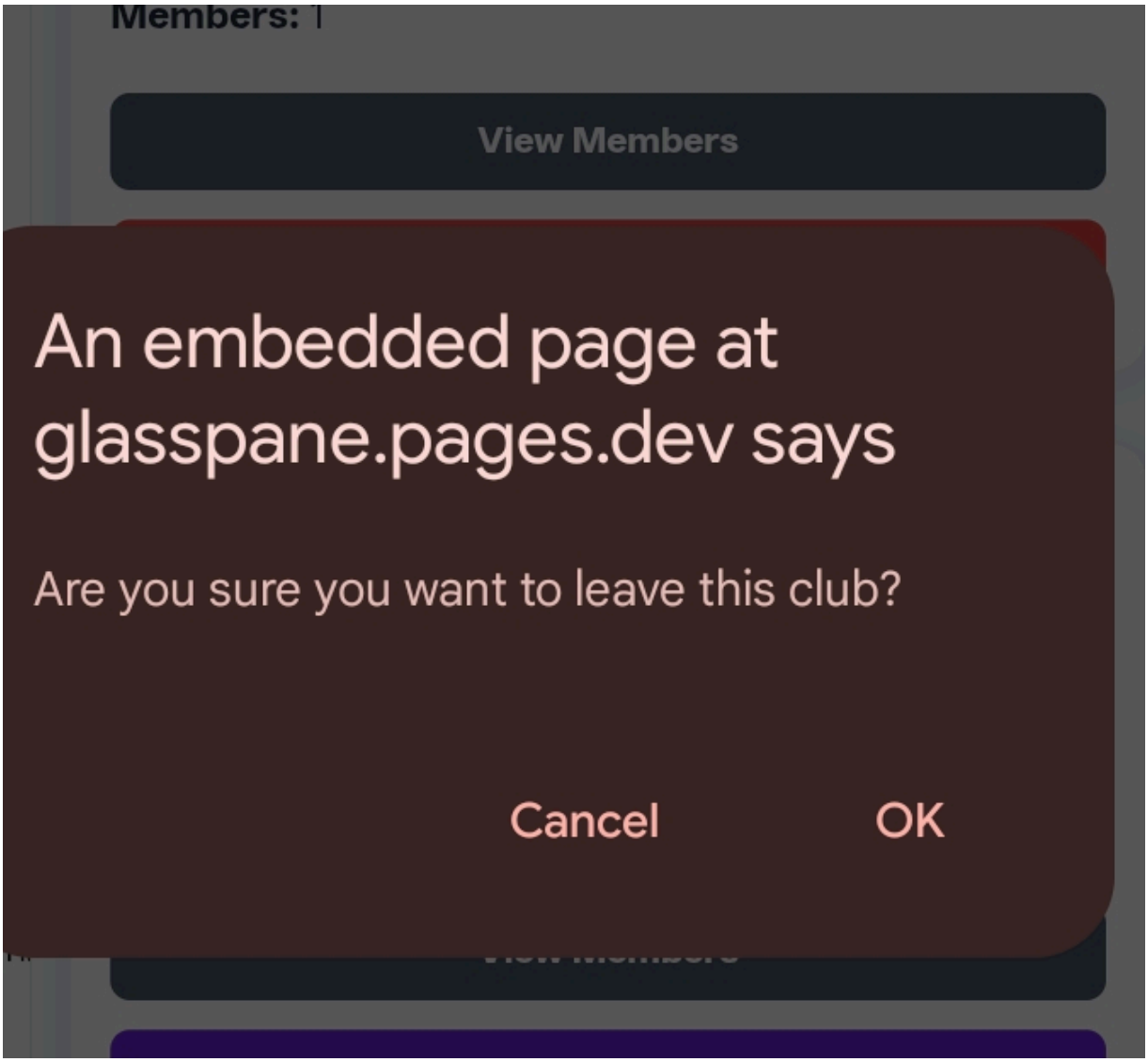
Step 3: Display Club Information



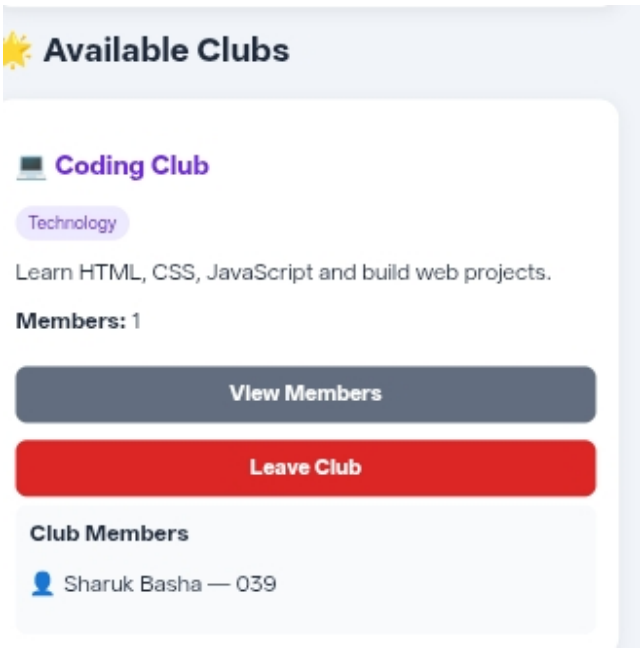
Step 4: Member Registration with Validation



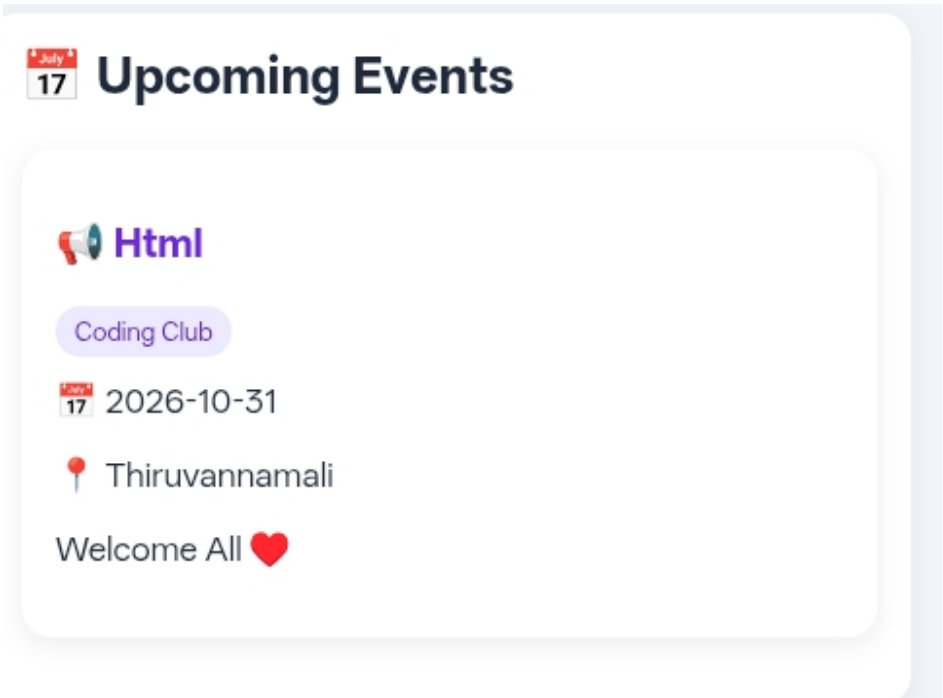
Step 5: Join and Leave Club



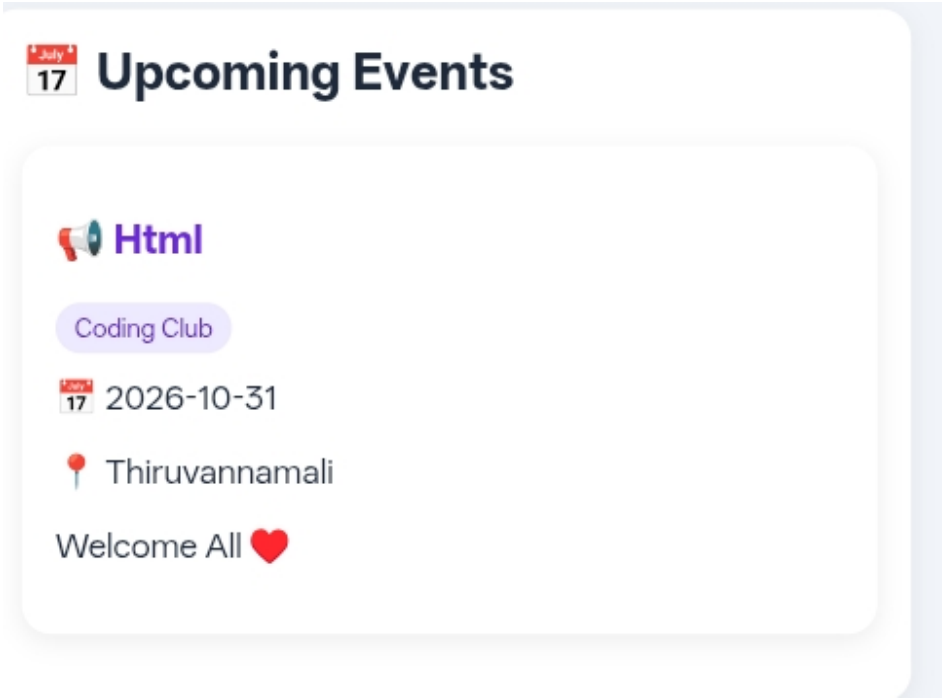
Step 6: Display Club Members



Step 7: Add Event Announcements



Step 8: Display Upcoming Club Events



Step 9: Store Membership Details Using Local Storage

🎓 Campus Club Hub

Discover Clubs | Join Communities | Explore Events

6

Available Clubs

3

Memberships

1

Upcoming Events

🔎 Explore Clubs

Search clubs...

All Categories

📝 Student Registration

Full Name

6. Technical Explanation

HTML: Creates the structure of the College Club Management System.

2. CSS: Designs the website using colors, layouts, and cards.

3. JavaScript: Handles club search, student registration, and event management.

7. Conclusion

The College Club Management System was developed using HTML, CSS, and JavaScript. It helps students search for clubs, register as members, and view upcoming events. Local Storage saves membership and event details. This project makes college club activities easy, organized, and user-friendly.

8. Future Enhancements

Add student and admin login.

Connect the website to a database.

Add online event registration.

Send notifications about upcoming events.

Improve mobile-friendly design.

Generate club member and event reports.